

Predicting Hotel Booking Cancellations

MSIS 502

Presented by: Team 4

**Eva Maheshwari, Hilton Nguyen, Kayla Pham,
Serena Mei, William Somat**



WHAT IS THE GOAL?



The Problem

- Roughly **1 in 3** hotel bookings gets cancelled before the guest arrives
- By the time the hotel finds out, it's usually too late to resell the room at a good rate

Our Question

Can we tell, at the moment a booking is made, whether it's **likely to be cancelled**?

Who this helps

Hotel managers deciding how much to overbook, which channels to invest in, and when to require a deposit

THE DATASET

A green car is driving on a curved road that starts from the top left and curves downwards to the bottom right. The road is dark gray with white dashed lines. The car is positioned in the middle of the curve, facing downwards.

Hotel Bookings Demand Data set

Kaggle dataset from a 2019 study by Antonio, Almeida & Nunes — one city hotel and one resort hotel in Portugal

Raw Numbers

119,390 Bookings

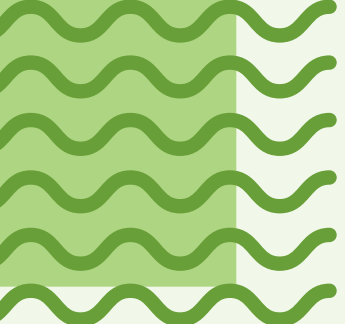
32 columns

2 Portuguese Hotels (City + Resort)

Across July 2015 to August 2017

Target Variable

Is_Canceled



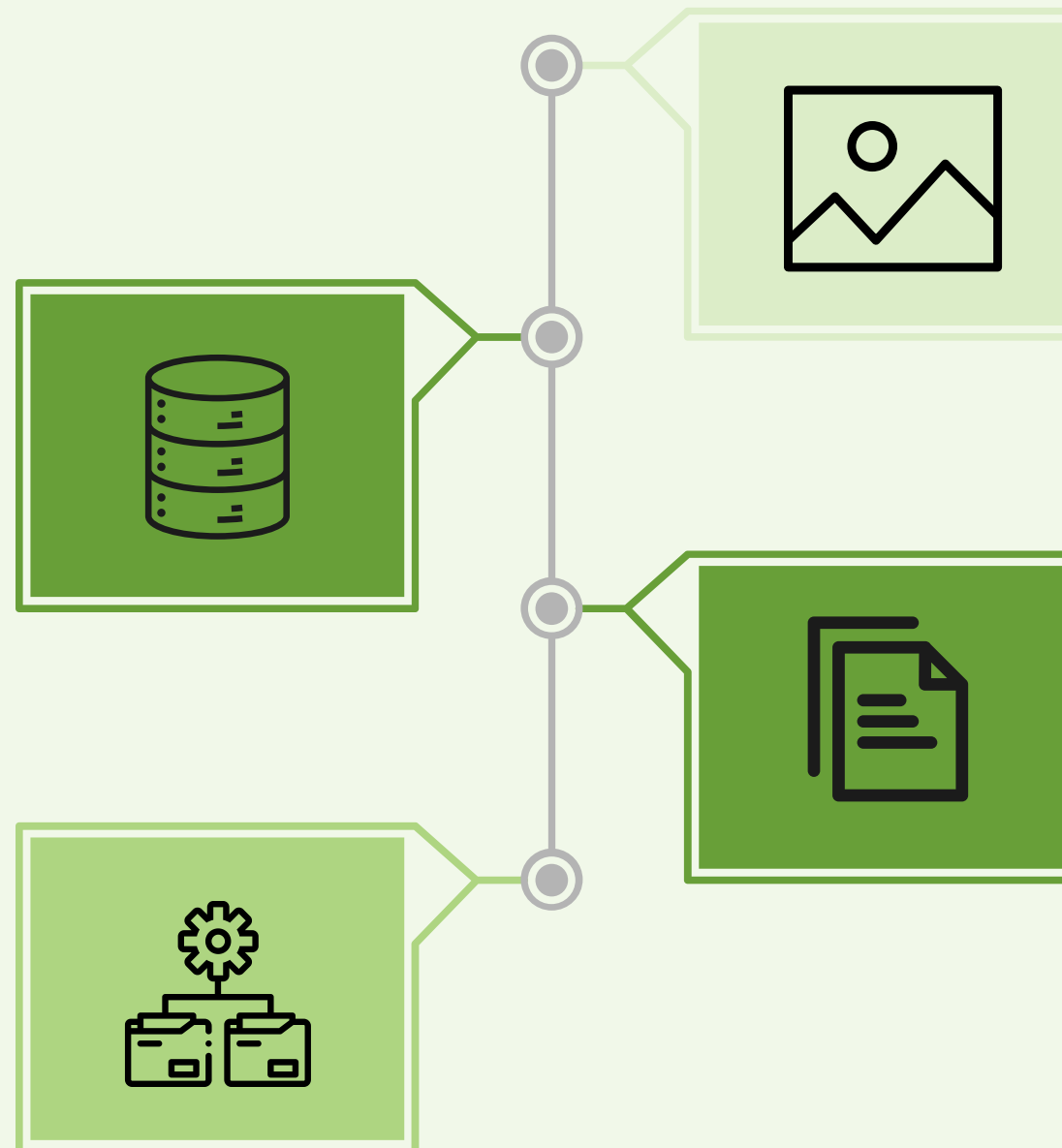
DATA CLEANING & PREPROCESSING

Invalid Values

- adr (nightly rate) ranged from **-6.38** to **5,400** → kept only bookings between **0–1,000**
- Removed bookings listing **0** adults (not a real reservation)

Class Balance

- **~37%** of bookings were cancelled, and **~63%** were not
- Balanced enough that a model has to actually learn—a model that always guessed "not cancelled" would still be right 63.2% of the time by doing nothing



Missing Values

- company: missing in **112,593** rows (94%) → column **dropped entirely**
- agent: missing in **16,340** rows (14%) → filled with **0** (no agent used)
- country: missing in **488** rows (<1%) → filled with **"Unknown"**
- children: missing in **4** rows → filled with **0**

Net Effect

- Removed **405** rows out of **119,390** (0.34%)
- Retained **99.7%** of original data → **118,985** bookings used going forward

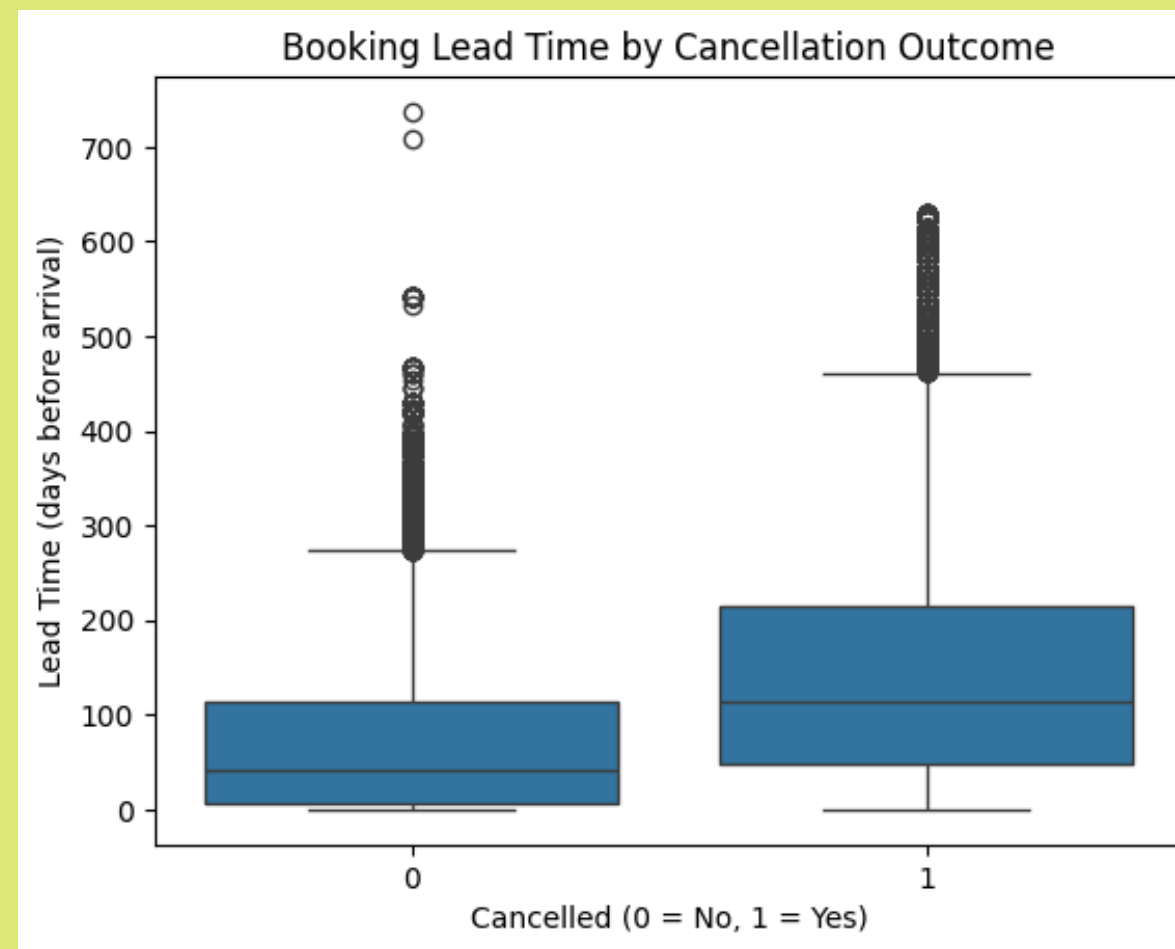
POTENTIAL DRIVERS OF HOTEL CANCELLATIONS

Property Type



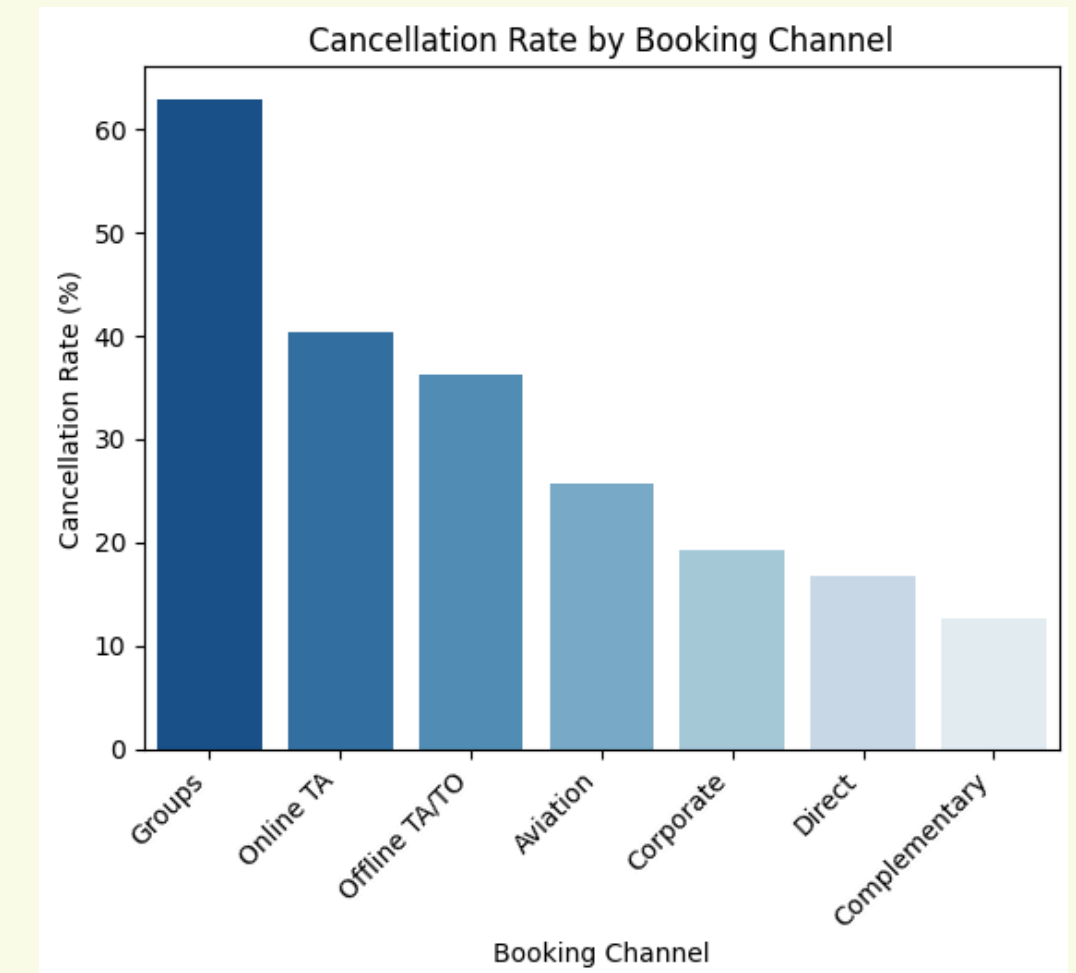
41.8% versus 27.8%

Days before Arrival



Booking that end up getting cancelled are much further in advance

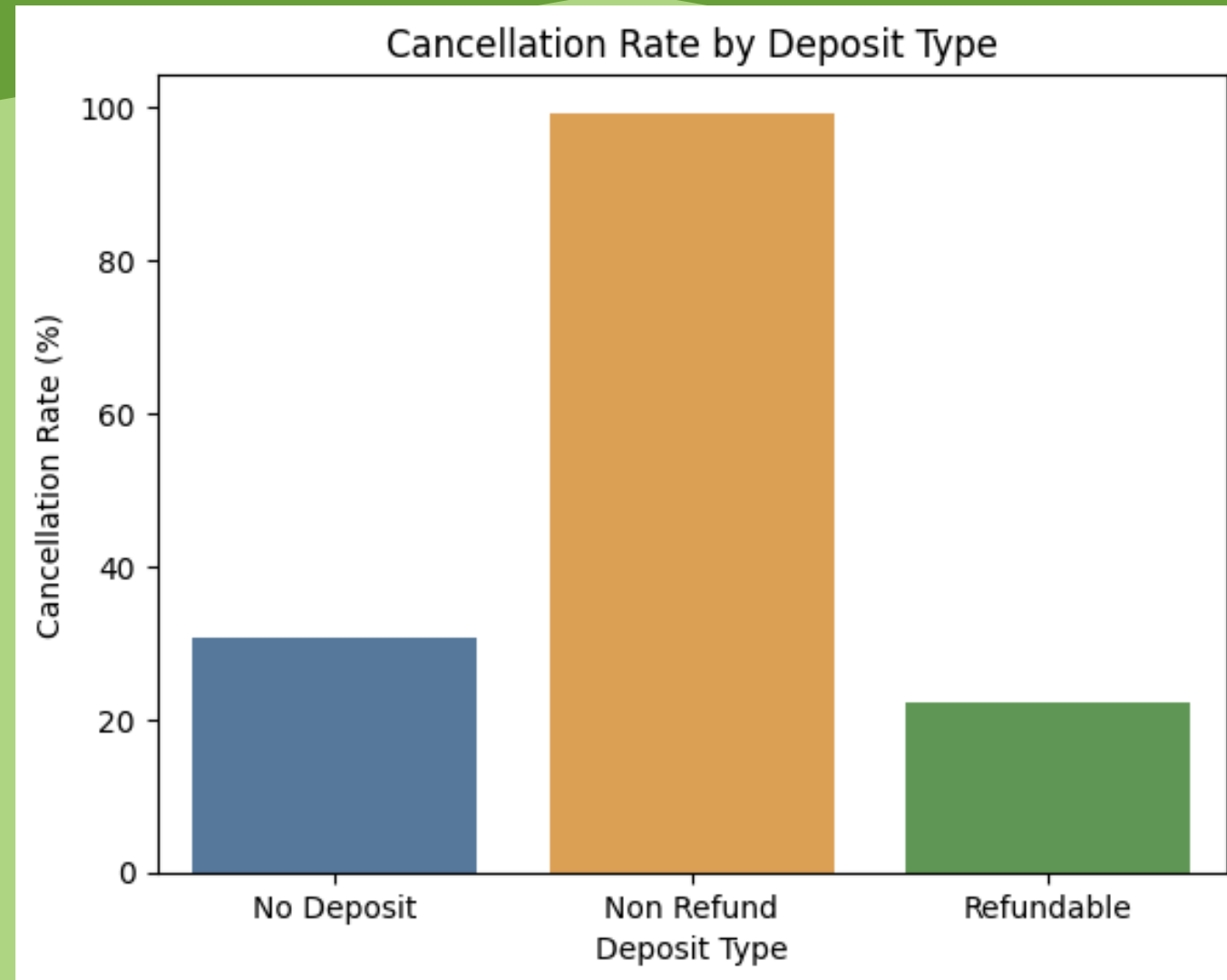
Resource



A direct booking is worth more to the hotel

POTENTIAL DRIVERS OF HOTEL CANCELLATIONS

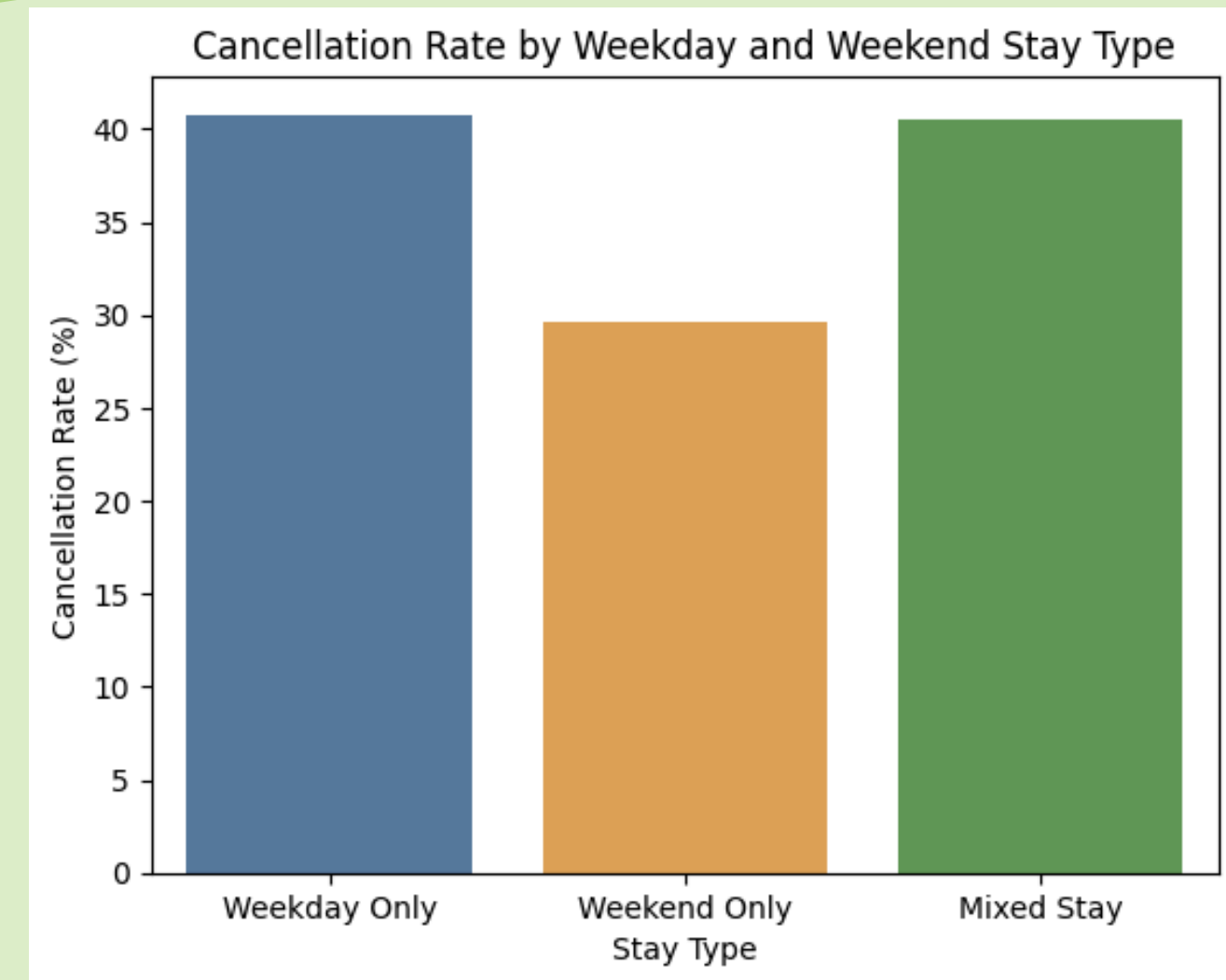
Deposit or Not



*Non-refundable: 12.3%, No Deposit: 87.6%

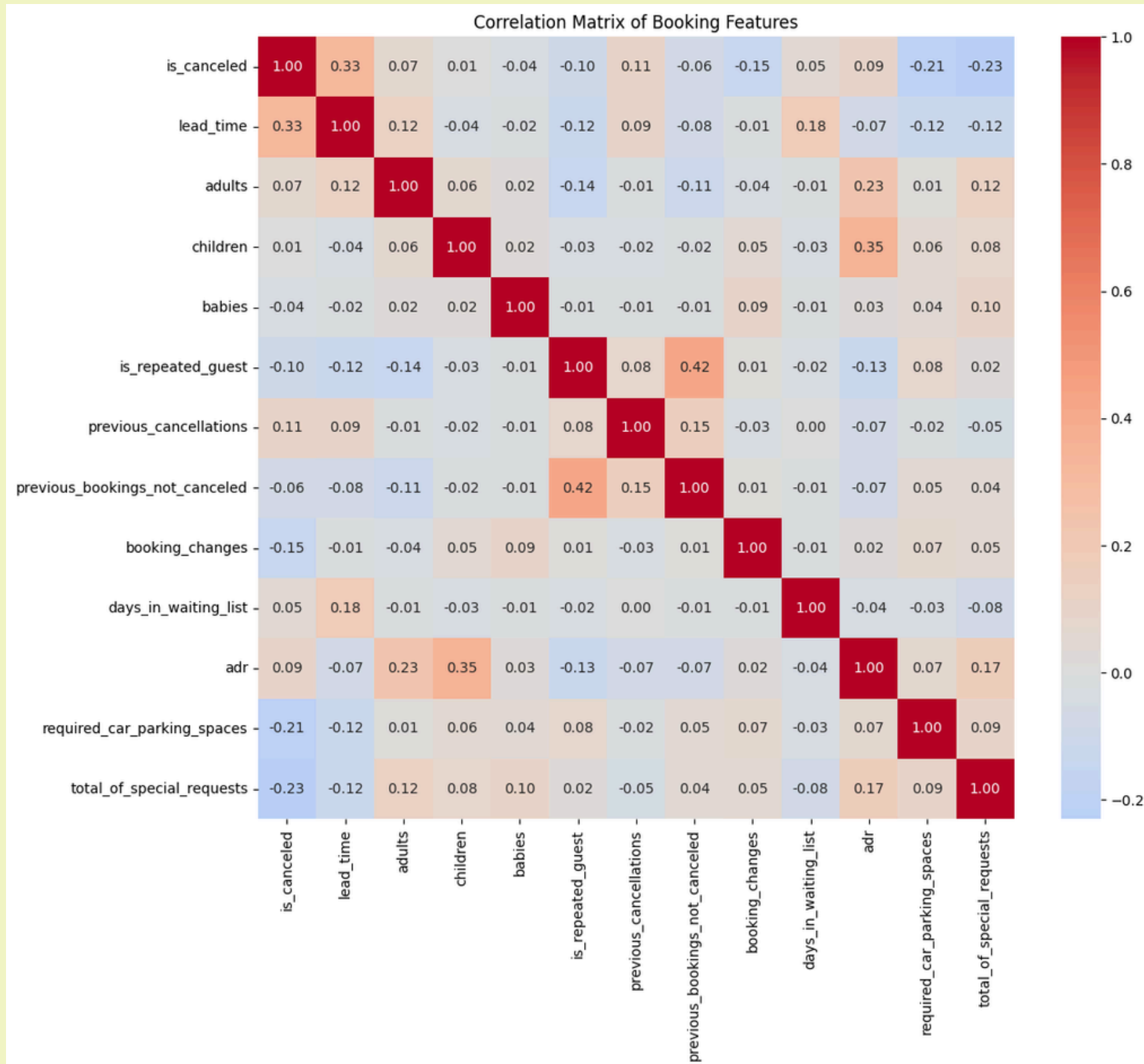
Despite the large differences in cancellation rates, the proportion of total bookings each category represents must also be considered

Weekday or Weekend

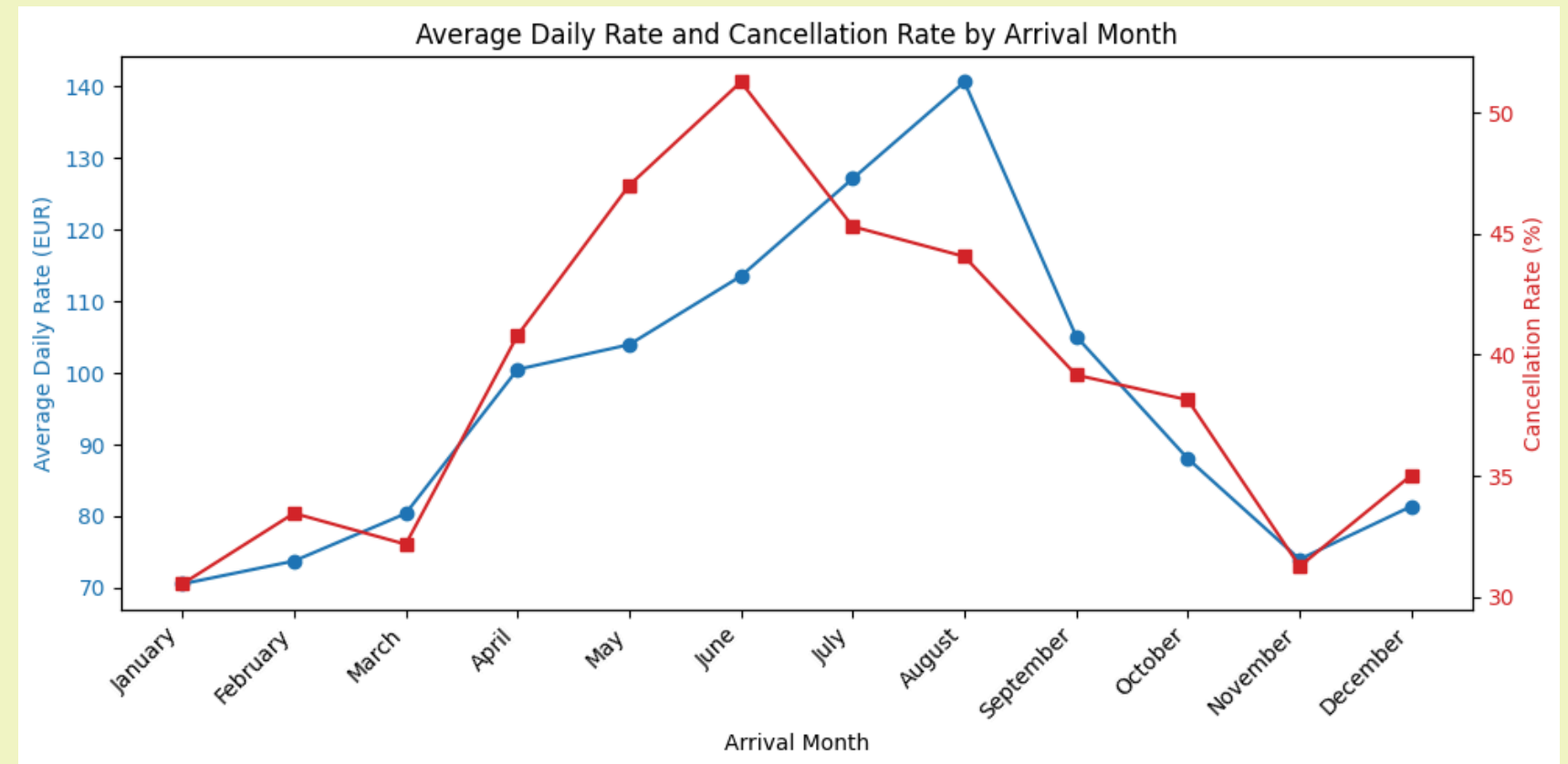


Weekend-only guests may have more definite travel plans. The hotel could use targeted reminders, deposits, or limited cancellation policies for weekday and mixed-stay reservations.

POTENTIAL DRIVERS OF HOTEL CANCELLATIONS



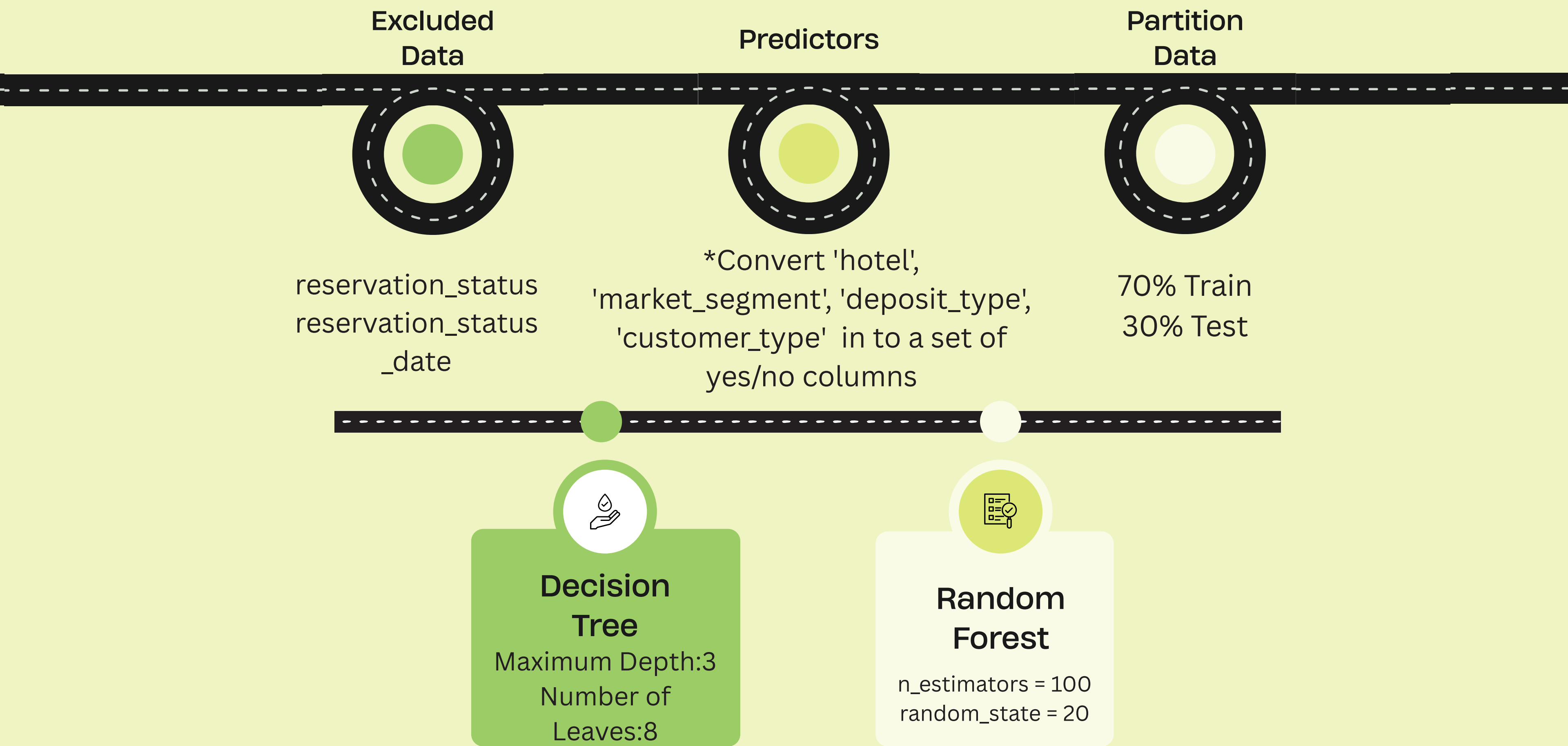
- **Lead time** has the strongest connection to cancellation (+0.33)
- The strongest negative connections are the **number of special requests** (-0.23) and whether the guest asked for a **parking space** (-0.21)



- Price varies: August(140.39) — January(70.52)
- Cancellations happens: June(41.5%), April(40.8%)
November(31.3%), January(30.5%)

Let's Build a Decision Tree!

PREDICTIVE MODELLING - FORESEEING CANCELLATIONS

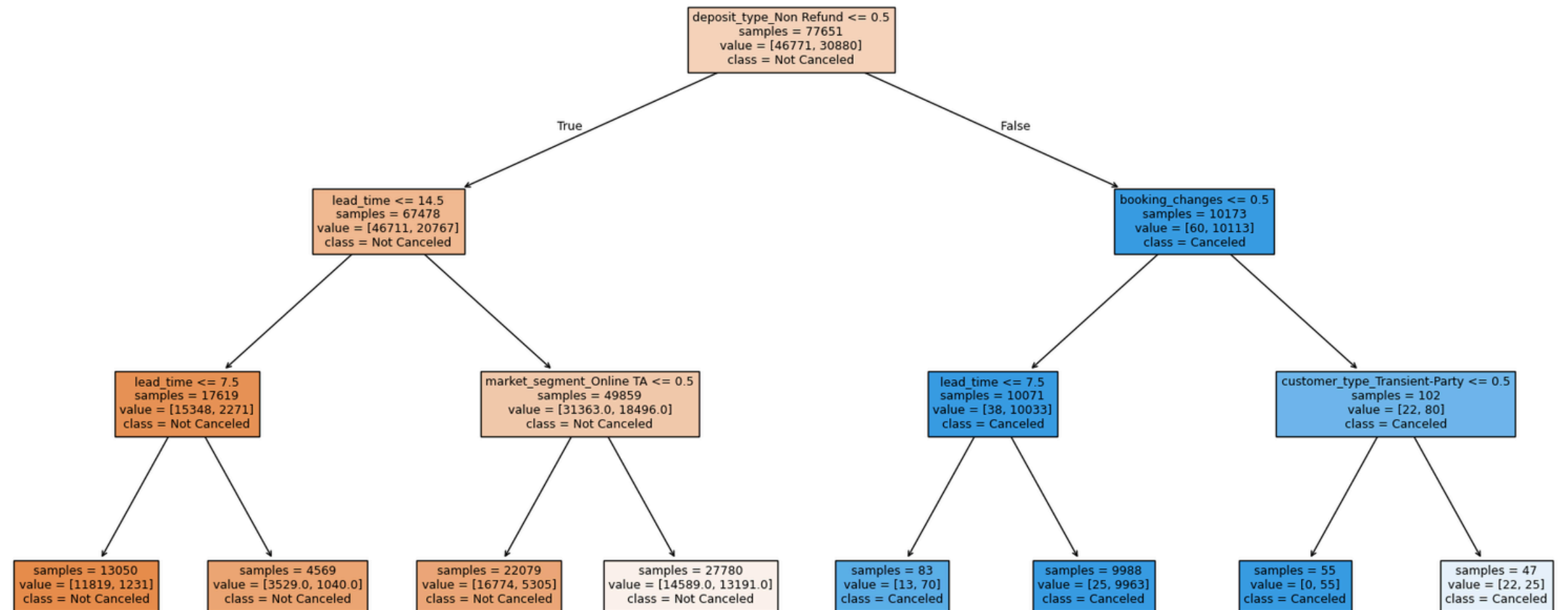


#1 CLASSIFICATION TREE

Non-refundable deposit

Lead time

Cancellation history



Baseline: 60.25%
Decision_tree_test: 73.31%
Decision_tree_train: 73.18%

#2 RANDOM FOREST REGRESSOR

TEST accuracy
84.96%

TRAIN accuracy
98.69%

Forest relies on...

Lead_time	0.2741
adr	0.2289
deposit_type_Non Refund	0.1245
total_of_special_requests	0.0631
stays_in_week_nights	0.0593

DOES ACCURACY TELL THE WHOLE STORY?



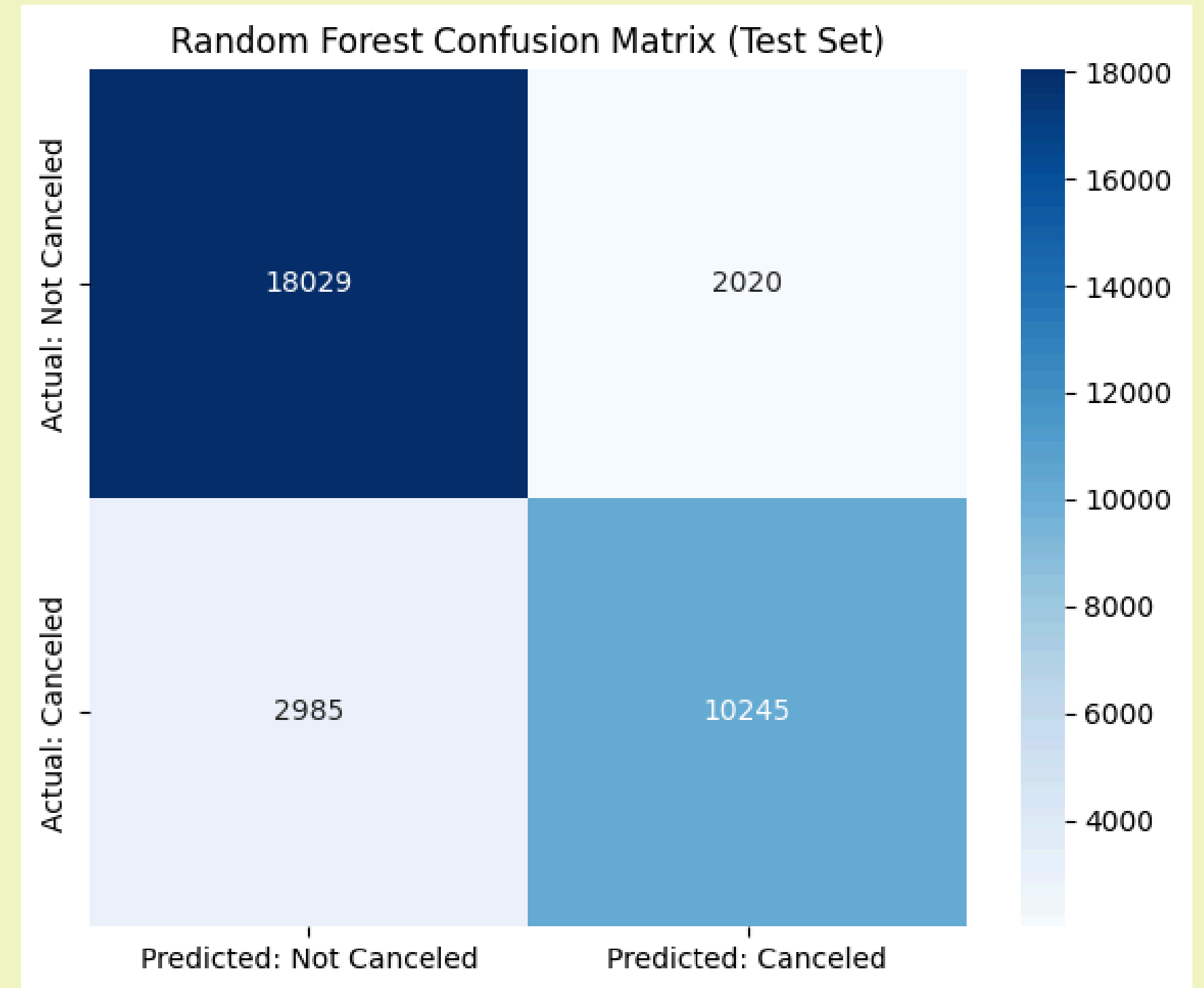
3,344 unpredicted cancellations

Hotel may plans around a guest who was never going to arrive.



2,245 unpredicted check-in

Guest may shows up with a unprepared room



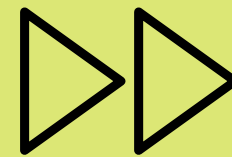
The hotel could adjust how cautious the model is depending on the season rather than leaving it on one setting all year.

BUSINESS INSIGHTS & RECOMMENDATIONS



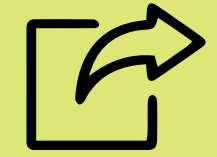
Key Findings

- **Lead time** is the strongest warning sign (113 vs. 45 days)
- **Booking source matters:** Direct 15.4% vs. Group 61.1% vs. Online TA 36.7%
- Guests who ask for anything (special request) are far **more likely** to show up
- **Peak season (Aug)** both costs more per room and cancels more often



Recommendations

- Set cancellation expectations by **channel + season**, not one flat rate
- Invest more in **direct-booking channels** (cheaper and far more reliable)
- Staff **based on expected arrivals** (~63% show rate), not raw booking counts
- Don't overreact to the 99.4% non-refundable rate — test it before changing deposit policy, since it may just reflect bookings the hotel already flagged as risky



Call to Action

- **Revenue managers** (overbooking & pricing decisions)
- **Marketing** (channel investment)
- **Operations** (staffing forecasts)

WHAT'S NEXT?

Limitations

- Data is 2015–2017, pre-pandemic, two hotels in Portugal only
- Parking column may be recorded at check-in, not booking (removing it only drops Model 2 accuracy from 84.3% to 83.9%)
- Cleaning involved judgment calls (adr cutoff at 1,000; kept 1,804 zero-rate bookings)

What We'd Do Differently

- Get more recent (post-pandemic) data
- Build a cost-sensitive model — a missed cancellation and a false alarm cost the hotel very different amounts
- Try predicting nightly rate (adr) as a regression task too

How We Used AI

- Used AI to structure brainstorming and catch errors in math/logic/code
- Verified all numbers and assumptions manually before including them